

Mathematics Bootcamp

Properties of Preferences and Utility Functions

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Today

- Preference Relations, Indifference, and Utility
- Monotonicity and Local Nonsatiation
- Convexity and Quasiconcavity
- Continuity Through Contour Sets
- Continuous Utility Representation

Preference Relations

- A **preference relation** $\succeq$ on X is a subset of $X \times X$.
- $x \succeq y$ reads: x is *at least as good* as y (x is *weakly better* than y).
- From $\succeq$ we derive:
 - Strict preference: $x \succ y \iff (x \succeq y \text{ and not } y \succeq x)$.
 - Indifference: $x \sim y \iff (x \succeq y \text{ and } y \succeq x)$.

Properties of Indifference

Indifference Relation

If $\succeq$ is rational (i.e., complete and transitive), then the induced indifference relation $\sim$ is:

1. **Reflexive:** $x \sim x$ for all $x \in X$.
2. **Symmetric:** $x \sim y \implies y \sim x$.
3. **Transitive:** $x \sim y \wedge y \sim z \implies x \sim z$.

Equivalence Relation

An equivalence relation is a binary relation that is reflexive, symmetric and transitive.

Equivalence Classes

Equivalence Classes

An equivalence relation induces a **partition** of the set: mutually disjoint subsets whose union equals the entire set.

If $\succeq$ is rational, its induced indifference relation $\sim$

1. is an equivalence relation;
2. generates equivalence classes of indifferent elements.

For each $x \in X$, the equivalence class of x is denoted by

$$[x] = \{y \in X \mid x \sim y\}.$$

Equivalence Classes and Utility

Utility representation

A function $u : X \rightarrow \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \iff u(x) \geq u(y).$$

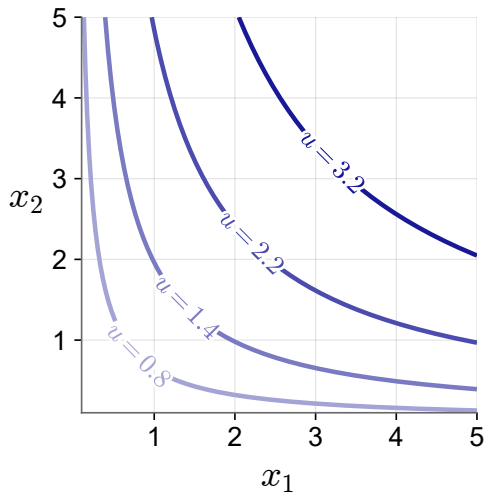
What the Representation Implies

If u represents $\succeq$, then

$$x \sim y \iff u(x) = u(y).$$

Every equivalence class maps to one utility level; utility is **ordinal**.

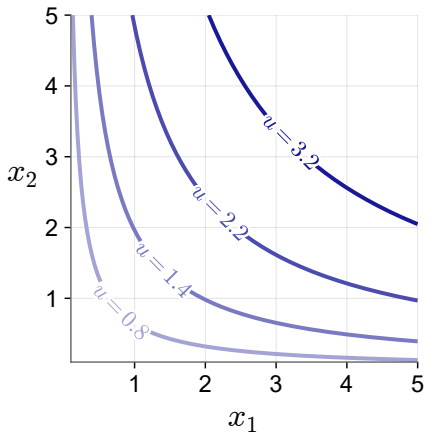
Indifference Curves



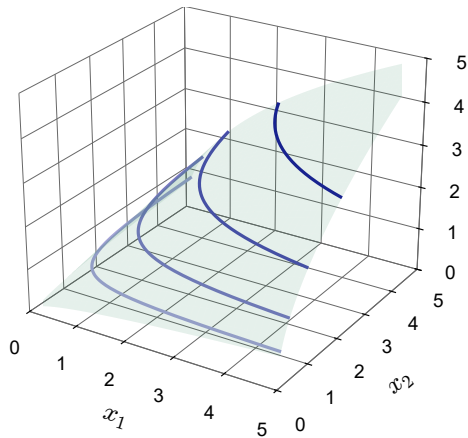
- Indifference curves are equivalence classes of X ; each maps to a single utility level in $\mathbb{R}$.
- Distinct equivalence classes are disjoint, so two indifference curves can never cross, or even touch.

Utility Representation

Indifference curves



Utility surface



Represent Commodities with Ordered Lists

2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to L (indexed by $\ell = 1, \dots, L$).

As a general matter, a *commodity vector* (or *commodity bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the *commodity space*.¹

We can use commodity vectors to represent an individual's consumption levels. The ℓ th entry of the commodity vector stands for the amount of commodity ℓ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

- $x \geq y$ iff $x_\ell \geq y_\ell$ for every ℓ ; and $x \gg y$ iff $x_\ell > y_\ell$ for every ℓ .
- So $x \geq y$ with $x \neq y$ means $x_\ell > y_\ell$ for *some* ℓ — weaker than $x \gg y$.

Monotonicity: More is Better

MWG 3.B.2: Monotonicity

Assume consuming more is always feasible: $x \in X$ and $y \geq x$ imply $y \in X$. The preference relation $\succeq$ on X is

monotone if $y \gg x$ implies $y \succ x$;

strongly monotone if $y \geq x$ and $y \neq x$ imply $y \succ x$.

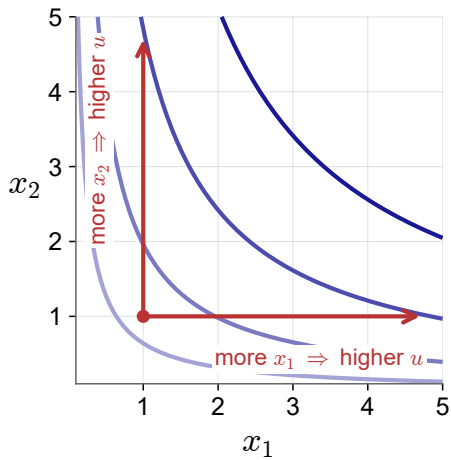
Proposition

If $u : X \rightarrow \mathbb{R}$ represents $\succeq$, then

$\succeq$ is monotone *iff* $u(y) > u(x)$ whenever $y \gg x$;

$\succeq$ is strongly monotone *iff* u is *strictly increasing*: $u(y) > u(x)$ whenever $y \geq x$ and $y \neq x$.

Monotonicity: More is Better



Domain: $X = \mathbb{R}_{++}^2$.

Local Nonsatiation: No Thick Indifference Curves

Definition: Local Nonsatiation

A preference relation $\succeq$ on X is *locally nonsatiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that $\|x - y\| \leq \varepsilon$ and $y \succ x$.

$$\text{Euclidean distance: } \|x - y\| = \sqrt{\sum_{i=1}^L (x_i - y_i)^2}.$$

Local Nonsatiation: No Thick Indifference Curves

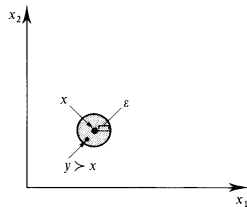


Figure 3.B.1

The test for local nonsatiation.

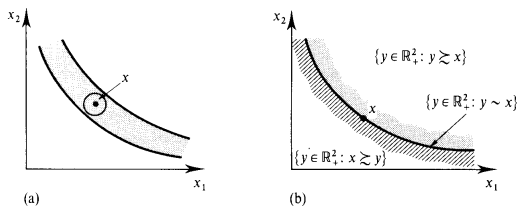


Figure 3.B.2

(a) A thick indifference set violates local nonsatiation.
 (b) Preferences compatible with local nonsatiation.

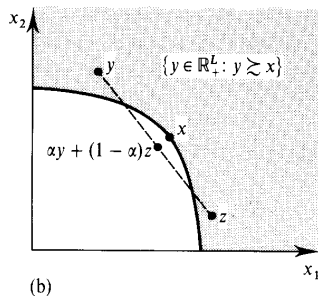
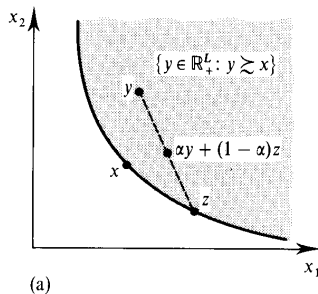
Monotonicity and Local Nonsatiation

Proposition

On $X = \mathbb{R}_+^L$, monotonicity implies local nonsatiation.

However, local nonsatiation does not imply monotonicity.

Convexity: Taste for Variety



Definition 3.B.4: The preference relation $\succsim$ on X is *convex* if for every $x \in X$, the upper contour set $\{y \in X: y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1 - \alpha)z \succsim x$ for any $\alpha \in [0, 1]$.

Figure 3.B.3(a) depicts a convex upper contour set; Figure 3.B.3(b) shows an upper contour set that is not convex.

Convexity: Taste for Variety

Convex Sets and Preferences

A set $X \subseteq \mathbb{R}^L$ is *convex* if $x, y \in X$ and $\alpha \in [0, 1]$ imply $\alpha x + (1 - \alpha)y \in X$.

Let X be convex and $U(x) := \{y \in X : y \succeq x\}$. Then $\succeq$ is *convex* if

$$y, z \in U(x), \alpha \in [0, 1] \implies \alpha y + (1 - \alpha)z \in U(x),$$

and *strictly convex* if that conclusion is $\succ x$ for $y \neq z, \alpha \in (0, 1)$.

Every upper contour set $U(x)$ is convex.

Level Sets and Quasiconcavity

Definitions

Let $X \subseteq \mathbb{R}^L$ be convex. A function $f : X \rightarrow \mathbb{R}$ is *quasiconcave* if, for $x, y \in X$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \geq \min\{f(x), f(y)\}.$$

It is *strictly quasiconcave* if, for $x \neq y$ and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)y) > \min\{f(x), f(y)\}.$$

Equivalently. f is quasiconcave iff every upper level set $\{x \in X \mid f(x) \geq c\}$ is convex.

Convex Preferences and Quasiconcave Utility

Kreps 2.8(b)

Let X be convex and let $u : X \rightarrow \mathbb{R}$ represent $\succeq$. Then

$$\begin{aligned}\succeq \text{ is convex} &\iff u \text{ is quasiconcave,} \\ \succeq \text{ is strictly convex} &\iff u \text{ is strictly quasiconcave.}\end{aligned}$$

Properties of a Representation

Property of $\succsim$	Property of a representation u
Monotonicity	$u(y) > u(x)$ whenever $y \gg x$.
Strong monotonicity	u is strictly increasing.
Convexity	u is quasiconcave.
Strict convexity	u is strictly quasiconcave.

Continuity is different: that is Debreu's theorem.

Topology on the Choice Set

Relative Topology on $X \subseteq \mathbb{R}^L$

For $x \in X$ and $\varepsilon > 0$, the relative open ball is

$$B_\varepsilon^X(x) = \{y \in X : \|y - x\| < \varepsilon\}.$$

- A set $G \subseteq X$ is *open relative to X* if every $x \in G$ has some $\varepsilon > 0$ with $B_\varepsilon^X(x) \subseteq G$.
- A set $F \subseteq X$ is *closed relative to X* if $X \setminus F$ is open relative to X .

Day 2 defined open, closed and bounded in $\mathbb{R}^L$. The same definitions, with the balls cut down to X , give the relative notions.

Closed Sets Contain Their Limits

Sequential Test for Closed Sets

Let $F \subseteq X \subseteq \mathbb{R}^L$. The set F is closed relative to X if and only if

$$x_n \in F \text{ for every } n, \quad x_n \rightarrow x \in X \quad \implies \quad x \in F.$$

Lower and Upper Contour Sets

Contour Sets at x

$$U(x) := \{y \in X \mid y \succeq x\} \quad L(x) := \{y \in X \mid x \succeq y\}.$$

The indifference set through x is always

$$I(x) := \{y \in X : y \sim x\} = U(x) \cap L(x).$$

Weak and Strict Sections

If $\succeq$ is complete, then

$$\{y \in X : y \succ x\} = X \setminus L(x), \quad \{y \in X : x \succ y\} = X \setminus U(x).$$

Lexicographic Preferences

Example 3.C.1: The Lexicographic Preference Relation

Let $X = \mathbb{R}_+^2$. Define $x \succeq y$ if either

$$x_1 > y_1, \quad \text{or} \quad x_1 = y_1 \text{ and } x_2 \geq y_2.$$

It is complete, transitive, strongly monotone, and strictly convex — yet no utility function represents it.

Continuity: Limits of Rankings

MWG 3.C.1: Continuity

Let $X \subseteq \mathbb{R}^L$ and let $\succsim$ be complete. Then $\succsim$ is *continuous* if it is preserved under limits:

$$(x_n, y_n) \rightarrow (x, y), \quad x_n \succsim y_n \text{ for every } n \quad \implies \quad x \succsim y.$$

Continuity: Equivalent Tests

Kreps 1.14: Equivalent Characterizations

For complete, transitive $\succeq$, continuity is equivalent to each of:

1. $U(x)$ and $L(x)$ are closed relative to X , for every x .
2. $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open relative to X , for every x .
3. If $x \succ y$, there is $\varepsilon > 0$ with $x' \succ y'$ whenever $\|x' - x\| < \varepsilon$ and $\|y' - y\| < \varepsilon$.

Kreps takes (3) as the definition; MWG takes the limit form.

Continuous Representations and Closed Contours

Closed Level Sets

If $f : X \rightarrow \mathbb{R}$ is continuous, then for every $c \in \mathbb{R}$ the sets

$$f^{-1}([c, \infty)) \quad \text{and} \quad f^{-1}((-\infty, c])$$

are closed relative to X .

A continuous representation therefore implies continuous preferences.

Debreu's Representation Theorem

Kreps 2.9: Debreu's Theorem

Let $X = \mathbb{R}_+^L$. If $\succeq$ is complete, transitive, and continuous on X , then there exists a *continuous* utility function $u : X \rightarrow \mathbb{R}$ such that

$$x \succeq y \iff u(x) \geq u(y).$$

- It gives *at least one* continuous representation — not every representation is continuous.

Three Assumptions, Three Conclusions

Assumptions	Conclusion
Nonempty compact D ; continuous u	$\arg \max_{x \in D} u(x)$ is nonempty.
Convex D ; strictly quasiconcave u	There is at most one maximizer.
Local nonsatiation; $D = B(p, I)$ with $p \gg 0$	Every optimum satisfies $p \cdot x = I$.

Day 4 takes each in turn.